Summary: Probability for machine learning

Joint distribution: $p_D(x_{\text{obs}}, z, y_*)$

Summarize
$E[g(X)] = \int g(x)p(x)\,\mathrm{d}x$

Marginalize
$p(x) = \int p(x, y)\,\mathrm{d}y$

Condition
$p(x|y) = \frac{p(x,y)}{p(y)}$

TAKE HOME

- The task determines which quantity we need from the joint distribution
- A small set of recurring operations—summarize, marginalize, condition—answers many of those questions
- Integrals are a central tool for answering these queries

Section 5

Joint Gaussian distributions and geometry

Describe and transform joint uncertainty with a mean and covariance

Gaussians: a concrete example

One variable

$$\mathcal{N}(x \mid \mu, \sigma^2) = \frac{\exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right)}{\sqrt{2\pi\sigma^2}}$$

***D* coordinates**

$$\mathcal{N}(\boldsymbol{x} \mid \mu, \Sigma) = \frac{\exp\left(-\frac{1}{2}(\boldsymbol{x}-\mu)^T\Sigma^{-1}(\boldsymbol{x}-\mu)\right)}{(2\pi)^{\frac{D}{2}}\ |\Sigma|^{\frac{1}{2}}}$$

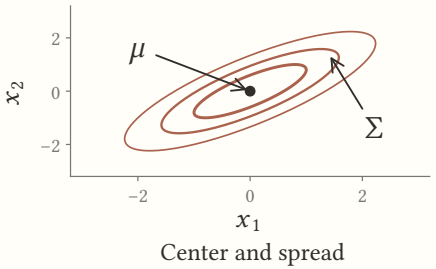
Murphy, Book 2 §2.3.1.1, Eq. 2.71; scalar case: $D = 1$

- One variable: mean μ and positive variance $\sigma^2 > 0$
- Mean $\mu \in \mathbb{R}^D$; these densities and inverse formulas assume covariance $\Sigma \in \mathbb{R}^{D\times D}$ with $\Sigma \succ 0$
- Gaussian means, marginals, and conditionals have closed forms

The three operations in multivariate Gaussian systems

One joint Gaussian: $p(\boldsymbol{x}) = \mathcal{N}(\boldsymbol{x} \mid \mu, \Sigma)$

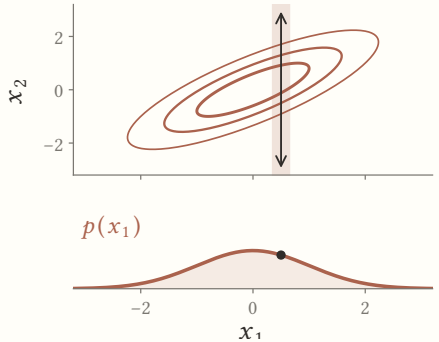
Summarize



Center and spread

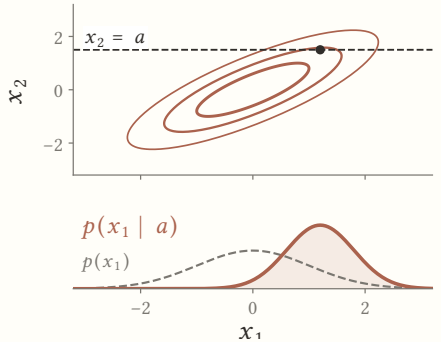
$$\mu = \int \boldsymbol{x} p(\boldsymbol{x}) \, \mathrm{d}\boldsymbol{x}$$

Marginalize



$$p(x_1) = \int p(x_1, x_2) \, \mathrm{d}x_2$$

Condition



$$p(x_1 \mid x_2 = a) = \frac{p(x_1, a)}{\int p(u, a) \, \mathrm{d}u}$$

Course figures generated by `gaussian_geometry_figures.ipynb`

- **Summarize:** describe the center and spread with μ and Σ
- **Marginalize:** integrate over the coordinate we ignore
- **Condition:** fix an observed coordinate and normalize the slice

Covariance summarizes joint variation

For a vector $\boldsymbol{x}$ with mean $\mu = E[\boldsymbol{x}]$

$$\Sigma_{ij} = \operatorname{Cov}[X_i, X_j]$$
$$= E[(X_i - \mu_i)(X_j - \mu_j)]$$

$$\Sigma = E[(\boldsymbol{x} - \mu)(\boldsymbol{x} - \mu)^T]$$
$$= E[\boldsymbol{x}\boldsymbol{x}^T] - \mu\mu^T$$

For three coordinates

$$\Sigma = \begin{bmatrix} V[X_1] & \operatorname{Cov}[X_1, X_2] & \operatorname{Cov}[X_1, X_3] \\ \operatorname{Cov}[X_2, X_1] & V[X_2] & \operatorname{Cov}[X_2, X_3] \\ \operatorname{Cov}[X_3, X_1] & \operatorname{Cov}[X_3, X_2] & V[X_3] \end{bmatrix}$$

Murphy, Book 1 §3.1.1, Eqs. 3.1–3.4

- Positive / negative covariance: deviations tend to have the same / opposite signs
- Diagonal entries are variances; off-diagonal entries are covariances
- Zero covariance does not imply independence

Correlation separates direction from units

$$\rho_{ij} = \frac{\mathrm{Cov}[X_i, X_j]}{\sigma_i \sigma_j}, \quad -1 \le \rho_{ij} \le 1$$

$$\Sigma = \begin{bmatrix} \sigma_1^2 & \rho_{12}\sigma_1\sigma_2 & \rho_{13}\sigma_1\sigma_3 \\ \rho_{12}\sigma_1\sigma_2 & \sigma_2^2 & \rho_{23}\sigma_2\sigma_3 \\ \rho_{13}\sigma_1\sigma_3 & \rho_{23}\sigma_2\sigma_3 & \sigma_3^2 \end{bmatrix}$$

Murphy, Book 1 §3.1.2, Eq. 3.7; Book 2 §2.3.1.1, Eq. 2.73

- ρ_{ij} describes the direction and strength of linear association
- $\sigma_1, \sigma_2, \sigma_3$ set the coordinate scales

Affine maps carry Gaussian means and covariances

$$\boldsymbol{y} = A\boldsymbol{x} + b$$

$$E[\boldsymbol{y}] = A\mu + b$$

$$\mathrm{Cov}[\boldsymbol{y}] = A\Sigma A^T$$

Covariance proof

Center: $\boldsymbol{d} = \boldsymbol{x} - \mu$

$$\begin{aligned}
\boldsymbol{y} &= A(\boldsymbol{d} + \mu) + b \\
&= A\boldsymbol{d} + (A\mu + b) \\
\boldsymbol{y} - E[\boldsymbol{y}] &= A\boldsymbol{d} \\
\mathrm{Cov}[\boldsymbol{y}] &= E\big[(A\boldsymbol{d})(A\boldsymbol{d})^T\big] \\
&= AE\big[\boldsymbol{d}\boldsymbol{d}^T\big]A^T \\
&= A\Sigma A^T
\end{aligned}$$

Murphy, Book 1 §3.1.1, Eq. 3.5

- Gaussianity is preserved; a shift changes only the mean
- A linear map changes the scale, orientation, and dimension of variation

Mahalanobis distance uses the cloud's own scale

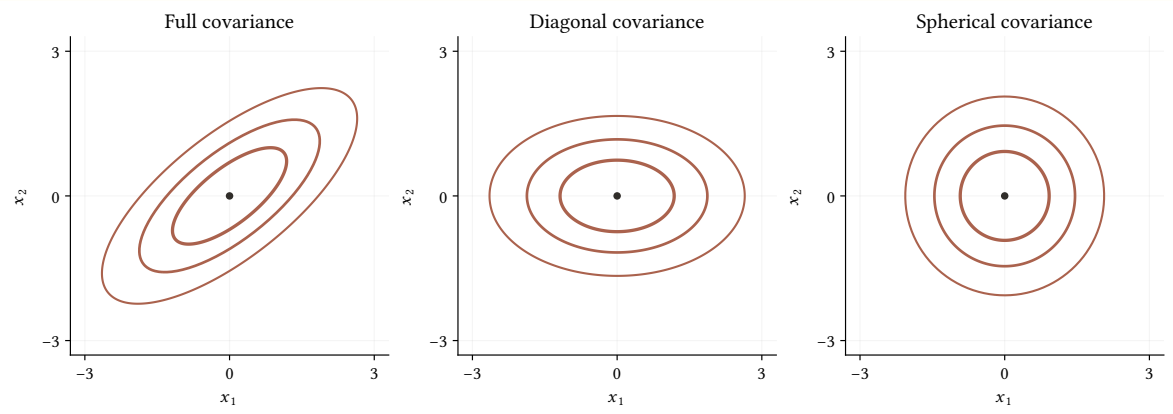
$$d_\Sigma(\boldsymbol{x}, \mu)^2 = (\boldsymbol{x} - \mu)^T \Sigma^{-1} (\boldsymbol{x} - \mu)$$

$$\Sigma = \sigma^2 I_D \quad \rightarrow \quad d_\Sigma^2 = \frac{\|\boldsymbol{x} - \mu\|^2}{\sigma^2}$$

Murphy, Book 2 §2.3.1.1, Eq. 2.72

- Equal Mahalanobis distances trace equal-density contours
- With spherical covariance, it becomes squared Euclidean distance in standard-deviation units

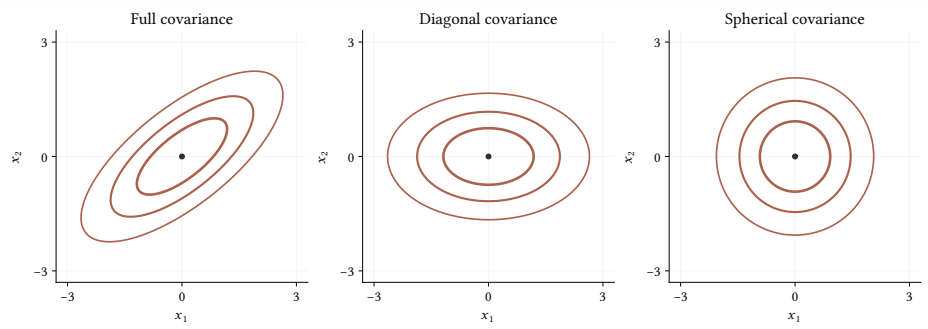
Covariance controls the contour shape



Course rendering of Murphy, Book 2, Fig. 2.8

- A full covariance can rotate the principal axes
- Diagonal covariance removes linear correlation in these coordinates
- Spherical covariance assigns the same variance in every direction

Summary: Joint Gaussian geometry



Affine map

$$\boldsymbol{y} = A\boldsymbol{x} + b$$
$$E[\boldsymbol{y}] = A\mu + b$$
$$\mathrm{Cov}[\boldsymbol{y}] = A\Sigma A^T$$

TAKE HOME

- Mean vector μ locates the cloud; covariance Σ sets its shape
- A linear map changes the scale, orientation, and dimension of variation

Section 6

Marginals and conditionals of a Gaussian

Ignore coordinates or use observations to update uncertainty

Partition the variables and their parameters together

$$\boldsymbol{x} = \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}, \quad \mu = \begin{pmatrix} \mu_1 \\ \mu_2 \end{pmatrix}$$

$$\Sigma = \begin{pmatrix} \Sigma_{11} & \Sigma_{12} \\ \Sigma_{21} & \Sigma_{22} \end{pmatrix}, \quad \Sigma_{21} = \Sigma_{12}^T$$

Murphy, Book 2 §2.3.1.3, Eq. 2.75

- x_1 and x_2 can each be vectors
- The diagonal blocks describe within-group variation
- The cross blocks describe variation between groups

Marginalization keeps the matching blocks

$$x \sim \mathcal{N}(\mu, \Sigma)$$

$$p(x_1) = \int p(x_1, x_2)\, \mathrm{d}x_2$$

$$= \mathcal{N}(x_1 \mid \mu_1, \Sigma_{11})$$

$$p(x_2) = \mathcal{N}(x_2 \mid \mu_2, \Sigma_{22})$$

Marginal proof

Select x_1: $P = [I \;\; 0]$, $\quad x_1 = Px$

By the affine rule:

$$x_1 \sim \mathcal{N}(P\mu, P\Sigma P^T)$$

$$P\mu = \mu_1$$

$$P\Sigma P^T = [I \;\; 0] \begin{bmatrix} \Sigma_{11} & \Sigma_{12} \\ \Sigma_{21} & \Sigma_{22} \end{bmatrix} \begin{bmatrix} I \\ 0 \end{bmatrix}$$

$$= \Sigma_{11}$$

Murphy, Book 2 §2.3.1.3, Eqs. 2.76–2.77; affine rule: Book 1 §3.1.1, Eq. 3.5

- Marginalizing x_2 integrates over all its possible values
- Cross-covariance affects conditioning, not the marginal of x_1

Conditioning shifts the mean by an observed deviation

$$p(x_1 \mid x_2) = \mathcal{N}\left(x_1 \mid \mu_{1|2}, \Sigma_{1|2}\right)$$

$$\mu_{1|2} = \mu_1 + \Sigma_{12}\Sigma_{22}^{-1}(x_2 - \mu_2)$$

Mean proof

$$B = \Sigma_{12}\Sigma_{22}^{-1}$$

$$r = x_1 - \mu_{1|2} = x_1 - \mu_1 - B(x_2 - \mu_2)$$

$$E[r] = 0,$$

$$\mathrm{Cov}[r, x_2] = \Sigma_{12} - B\Sigma_{22}$$

$$= 0$$

(r, x_2) is jointly Gaussian:

zero cross-covariance gives independence

$$E[r \mid x_2] = E[r] = 0$$

$$E[x_1 \mid x_2] = \mu_{1|2}$$

Murphy, Book 2 §2.3.1.3, Eq. 2.78; proof uses affine Gaussian closure

- Conditioning shifts the mean according to the observed deviation $x_2 - \mu_2$, cross-covariance, and observed variance

Conditioning removes explained covariance

$$\Sigma_{1|2} = \Sigma_{11} - \Sigma_{12}\Sigma_{22}^{-1}\Sigma_{21}$$

$$\Sigma_{2|1} = \Sigma_{22} - \Sigma_{21}\Sigma_{11}^{-1}\Sigma_{12}$$

Covariance proof

$$B = \Sigma_{12}\Sigma_{22}^{-1}$$

$$r = x_1 - \mu_{1|2}$$

From the mean proof: r and x_2 are independent

Given x_2, the mean $\mu_{1|2}$ is constant

$$\Sigma_{1|2} = \mathrm{Cov}[x_1 \mid x_2] = \mathrm{Cov}[r]$$

$$\Sigma_{11} = B\Sigma_{22}B^T + \mathrm{Cov}[r]$$

$$\Sigma_{1|2} = \Sigma_{11} - B\Sigma_{22}B^T$$

$$= \Sigma_{11} - \Sigma_{12}\Sigma_{22}^{-1}\Sigma_{21}$$

Murphy, Book 2 §2.3.1.3, Eqs. 2.78–2.79; proof reuses the independent residual

- Conditioning cannot increase variance in any direction
- In a Gaussian, this covariance depends on which variables are observed, not their values

The bivariate formula gives intuition

$$x_1, x_2 \in \mathbb{R}$$

$$(X_1 \mid X_2 = x_2) \sim \mathcal{N}\left(\mu_{1|2}, \sigma^2_{1|2}\right)$$

$$\mu_{1|2} = \mu_1 + \rho \frac{\sigma_1}{\sigma_2}(x_2 - \mu_2)$$

$$\sigma_{1|2} = \sigma_1 \sqrt{1 - \rho^2}$$

Murphy, Book 1 §3.2.4; Book 2 §2.3.1.3

- Positive ρ moves the estimate in the observed direction
- $\rho = 0$ leaves both mean and variance unchanged
- Larger $|\rho|$ leaves less conditional variance

One observed coordinate can change both the estimate and its uncertainty

Supplied joint model

$$\begin{pmatrix} X_1 \\ X_2 \end{pmatrix} \sim \mathcal{N}\left(\mu = \begin{pmatrix} 0 \\ 0 \end{pmatrix}, \Sigma = \begin{pmatrix} 1 & 0.8 \\ 0.8 & 1 \end{pmatrix}\right)$$

Recorded: $X_2 = 1.5$; **missing:** X_1

$X_2 = 1.5$ **recorded**

$$(X_1 \mid X_2 = 1.5) \sim \mathcal{N}(0.8(1.5), 1 - 0.8^2) = \mathcal{N}(1.2, 0.36)$$

X_2 **unobserved**

$$X_1 \sim \mathcal{N}(0, 1)$$

Course-constructed example; Gaussian conditioning: Murphy, Book 2 §2.3.1.3, Eq. 2.78

- With X_2 unobserved, marginalize it to obtain $p(X_1)$.
- Both calculations use supplied μ, Σ; neither fits them.
- Compared with the marginal, conditioning raises the mean to 1.2 and reduces the variance to 0.36.

The Schur complement is the residual covariance

Conditional Gaussian

$$\mu_{1|2} = \mu_1 + \Sigma_{12}\Sigma_{22}^{-1}(x_2 - \mu_2)$$
$$S = \Sigma_{1|2} = \Sigma_{11} - \Sigma_{12}\Sigma_{22}^{-1}\Sigma_{21}$$
$$(x_1 \mid x_2) \sim \mathcal{N}\big(\mu_{1|2}, S\big)$$

Centered residual

$$r = x_1 - \mu_{1|2}$$
$$r \sim \mathcal{N}(0, S)$$

The residual r is independent of x_2

Remove the part predicted by x_2:

$$\mathrm{Cov}\left[\begin{pmatrix} r \\ x_2 \end{pmatrix}\right] = \begin{pmatrix} S & 0 \\ 0 & \Sigma_{22} \end{pmatrix}$$

Murphy, Book 2 §2.3.1.5, Eqs. 2.91, 2.102–2.103

- S measures the uncertainty left after subtracting the conditional mean
- Block elimination leaves S: the **Schur complement of Σ_{22} in Σ**

The joint quadratic separates into conditional and marginal parts

$$p(x_1, x_2) = p(x_1 \mid x_2)p(x_2)$$

$$(\boldsymbol{x} - \mu)^T \Sigma^{-1} (\boldsymbol{x} - \mu)$$

$$= r^T S^{-1} r + (x_2 - \mu_2)^T \Sigma_{22}^{-1} (x_2 - \mu_2)$$

Murphy, Book 2 §2.3.1.5, Eqs. 2.94–2.103

- The first quadratic describes x_1 after seeing x_2
- The second describes the marginal distribution of x_2

Summary: Marginals and conditionals of a Gaussian

Marginalize

$$p(x_1) = \int p(x_1, x_2)\, \mathrm{d}x_2$$
$$x_1 \sim \mathcal{N}(\mu_1, \Sigma_{11})$$

Condition

$$p(x_1|x_2) = \mathcal{N}\left(x_1|\mu_{1|2}, \Sigma_{1|2}\right)$$
$$\mu_{1|2} = \mu_1 + \Sigma_{12}\Sigma_{22}^{-1}(x_2 - \mu_2)$$
$$\Sigma_{1|2} = \Sigma_{11} - \Sigma_{12}\Sigma_{22}^{-1}\Sigma_{21}$$

TAKE HOME

- Marginalizing x_2 integrates over all its possible values
- Conditioning shifts the mean according to the observed deviation, cross-covariance, and observed variance
- Conditioning cannot increase variance in any direction

One linear Gaussian system supports different questions

$$z \sim \mathcal{N}(\mu_z, \Sigma_z), \quad y \mid z \sim \mathcal{N}(Wz + b, \Omega)$$

Condition

$$p(z \mid y) = \frac{p(y \mid z)p(z)}{p(y)}$$

Summarize

$$E[g(z) \mid y] = \int g(z)p(z \mid y)\,\mathrm{d}z$$

Marginalize

$$p(y) = \int p(y \mid z)p(z)\,\mathrm{d}z$$

Murphy, Book 2 §2.3.2

- Conditioning asks what we can learn from an observed measurement
  - What is the distribution of the hidden state after seeing a measurement?
- Summarizing turns a distribution into a quantity
  - What is our estimate of the hidden state?
- Marginalization averages over an unknown variable to predict another
  - What is the distribution of a measurement before it is observed?

A linear Gaussian model defines a joint Gaussian

Linear Gaussian model

$$z \sim \mathcal{N}(\mu_z, \Sigma_z)$$
$$y = Wz + b + \varepsilon$$
$$\varepsilon \sim \mathcal{N}(0, \Omega)$$

ε is independent of z

Affine Gaussian rules

$$u = Ax + c$$
$$E[u] = AE[x] + c$$
$$\mathrm{Cov}[u] = A\ \mathrm{Cov}[x]A^T$$

Joint Gaussian

$$\begin{pmatrix} z \\ y \end{pmatrix} \sim \mathcal{N}\left(\begin{pmatrix} \mu_z \\ \mu_y \end{pmatrix}, \begin{pmatrix} \Sigma_{zz} & \Sigma_{zy} \\ \Sigma_{yz} & \Sigma_{yy} \end{pmatrix}\right)$$
$$\mu_y = W\mu_z + b, \quad \Sigma_{zz} = \Sigma_z$$
$$\Sigma_{zy} = \Sigma_z W^T = \Sigma_{yz}^T$$
$$\Sigma_{yy} = W\Sigma_z W^T + \Omega$$

Murphy, Book 2 §2.3.2.1, Eq. 2.120

- The model plus affine-Gaussian rules determine the full joint distribution
- Once $p(z, y)$ is known, the Gaussian marginal and conditioning rules apply directly

Bayes' rule for Gaussians

Model and joint Gaussian

$$\begin{pmatrix} z \\ y \end{pmatrix} \sim \mathcal{N}\left(\begin{pmatrix} \mu_z \\ \mu_y \end{pmatrix}, \begin{pmatrix} \Sigma_{zz} & \Sigma_{zy} \\ \Sigma_{yz} & \Sigma_{yy} \end{pmatrix} \right)$$

$$\mu_y = W\mu_z + b, \quad \Sigma_{zz} = \Sigma_z$$
$$\Sigma_{zy} = \Sigma_z W^T = \Sigma_{yz}^T$$
$$\Sigma_{yy} = W\Sigma_z W^T + \Omega$$

Gaussian conditioning rule

$$\mu_{1|2} = \mu_1 + \Sigma_{12}\Sigma_{22}^{-1}(x_2 - \mu_2)$$
$$\Sigma_{1|2} = \Sigma_{11} - \Sigma_{12}\Sigma_{22}^{-1}\Sigma_{21}$$

Apply: $x_1 = z, x_2 = y$

$$p(z \mid y) = \mathcal{N}\left(z \mid \mu_{z|y}, \Sigma_{z|y}\right)$$

$$\mu_{z|y} = \mu_z + \Sigma_{zy}\Sigma_{yy}^{-1}\left(y - \mu_y\right)$$

$$\Sigma_{z|y} = \Sigma_{zz} - \Sigma_{zy}\Sigma_{yy}^{-1}\Sigma_{yz}$$

Murphy, Book 2 §2.3.2.2, Eq. 2.121

- Apply the Gaussian conditioning rule to the joint blocks
- The mean responds to observed minus predicted measurement
- The covariance subtracts the uncertainty explained by the measurement

Bayes' rule as an update formula

Carried forward Earlier: Gaussian conditioning

$$\mu_{z|y} = \mu_z + \Sigma_{zy}\Sigma_{yy}^{-1}(y - \mu_y)$$
$$\mu_y = W\mu_z + b$$
$$\Sigma_{zy} = \Sigma_z W^T$$
$$\Sigma_{yy} = W\Sigma_z W^T + \Omega$$

Define the gain

$$K = \Sigma_{zy}\Sigma_{yy}^{-1}$$
$$\mu_{z|y} = \mu_z + K(y - \mu_y)$$

Murphy, Book 2 §2.3.2.2, Eqs. 2.122–2.124

- K is the gain: it converts a measurement discrepancy into a state correction
- Posterior estimate = prior estimate + gain × prediction error

The scalar Gaussian update

Carried forward Earlier: Gaussian update formula

$$\mu_{z|y} = \mu_z + K(y - \mu_y)$$
$$K = \Sigma_{zy}\Sigma_{yy}^{-1}$$

Scalar specialization

$$y, z \in \mathbb{R}, \quad W = 1, \quad b = 0$$
$$\Sigma_z = \sigma_z^2, \quad \Omega = \sigma_\varepsilon^2$$
$$z \sim \mathcal{N}(\mu_z, \sigma_z^2)$$
$$y = z + \varepsilon, \quad \varepsilon \sim \mathcal{N}(0, \sigma_\varepsilon^2)$$
ε is independent of z

Predicted mean

$$\mu_y = E[z + \varepsilon] = \mu_z$$

Cross-covariance

$$\Sigma_{zy} = \mathrm{Cov}[z, z + \varepsilon] = \sigma_z^2$$

Measurement variance

$$\Sigma_{yy} = \mathrm{Var}[z + \varepsilon] = \sigma_z^2 + \sigma_\varepsilon^2$$

$$K = \frac{\sigma_z^2}{\sigma_z^2 + \sigma_\varepsilon^2} \quad \rightarrow \quad \mu_{z|y} = \mu_z + K(y - \mu_z)$$

Murphy, Book 2 §2.3.2.2; scalar specialization of Eqs. 2.121–2.123

- The scalar gain is prior state variance divided by total measurement variance

The update is weighted by relative uncertainty

$$\mu_{z|y} = \mu_z + K(y - \mu_z), \quad K = \frac{\sigma_z^2}{\sigma_z^2 + \sigma_\varepsilon^2}$$

◆ = posterior mean $\mu_{z|y}$

Precise measurement

$\sigma_\varepsilon^2 \ll \sigma_z^2 \quad K \approx 1$



Trust the measurement

Equal uncertainty

$\sigma_\varepsilon^2 = \sigma_z^2 \quad K = \frac{1}{2}$



Split the difference

Noisy measurement

$\sigma_\varepsilon^2 \gg \sigma_z^2 \quad K \approx 0$



Trust the prior

Scalar interpretation of Murphy, Book 2 §2.3.2.2

- K is the fraction of the prediction error used in the correction
- More measurement noise gives smaller K; more prior uncertainty gives larger K

Inference often combines conditioning and summarizing

Observed y $\rightarrow$ **Condition** $p(z \mid y) = \mathcal{N}\left(z \mid \mu_{z|y}, \Sigma_{z|y}\right)$ $\rightarrow$ **Summarize** $\hat{z} = E[z \mid y] = \mu_{z|y}$

Murphy, Book 2 §2.3.2.2–2.3.2.3

$$E[z \mid y] = \int z p(z \mid y) \, \mathrm{d}z$$

- Conditioning preserves the whole posterior distribution and its uncertainty
- Summarizing extracts a quantity from that posterior; for a Gaussian the mean is $\mu_{z|y}$